

View Review

Paper ID

652

Paper Title

Cartoonification of the real image using machine learning

Track Name

Track3:Computing Technology

REVIEW QUESTIONS

1. Determine if the paper adheres to the IEEE formatAccept

2. Abstract: Make sure the abstract contains proposed name, objective and research findings

The goal, which is to use machine learning and image processing techniques to cartoonize real photos, is stated explicitly in the abstract. OpenCV stylization, adaptive thresholding, and K-Means clustering are all part of the process. Real-time cartoonification using a Flask-based interface is the study finding that is discussed. Although the concept could be more focused and succinct, it still satisfies expectations overall.

3. Introduction: Make sure the Introduction contains the problem statement and the contribution of the work

The idea of cartoonification and its application to computer vision are briefly discussed in the introduction. It describes the issue converting actual photographs into creative cartoon representations and the solution, which is to integrate this into a web application using lightweight, conventional techniques instead of deep learning. Although this part is fine, it might do a better job of highlighting the problem's importance and distinctive contribution.

4. Related Work/Literature Survey:This section may be included in the introduction too. Carefully check if minimum 4-6 related articles are referred from 2022-2025

At least ten of the 15 references in the paper are from 2022–2025. Citations are made to well-known IEEE conference papers, GAN-based models (CycleGAN, AnimeGAN), and conventional filtering techniques. Nevertheless, there is no critical comparison or thorough explanation of how this study builds upon or differs from earlier efforts in the literature review.

5. Methodology:This section should contain novelty. Carefully check if the dataset information's are cited properly or well explained, mathematical justification for the proposed method, architecture diagram/block diagram of the proposed method

The methodology is extensive and modular, covering:

- ☐ Various styles of cartooning

- ☐ Each method's mathematical formulation
- ☐ OpenCV plus Flask usage
- ☐ There is a clear and present architecture diagram.

However, reproducibility is limited because no particular dataset is referenced or benchmarked. Combining lightweight techniques and implementing them in real-time is innovative, although this is more of an engineering contribution than a fundamental research breakthrough.

6. Results: Carefully evaluate if the proposed method is compared with existing similar approaches and the results are presented well in the form of graphs and tables

This is the weakest part of the paper. Although a few visual outputs, such as cartoonized picture figures, are displayed, there are

- ☐ No quantitative comparison with other methods
- ☐ No user study, no graphs or tables
- ☐ No analysis of processing time, accuracy, or quality metrics

This section needs major improvements for acceptance.

7. Conclusion: Make sure the conclusion contains findings of the work and the future scope

The findings are succinctly summarized in the conclusion, which also highlights the viability and efficiency of the employed strategies. Future scope is also mentioned, including cloud deployment for high-resolution pictures, transfer learning, and GAN integration. The conclusion is sufficient.

8. References: Make sure a minimum of 10-15 recent papers

The reference list includes 15 references, with at least 10 from 2022–2025, including IEEE, journals, and conferences. Proper citation format is followed

9. Overall Comments: Please provide your overall comments to the author

A well-organized system for real-time cartoonification utilizing web deployment and conventional machine learning approaches is presented in the article. Its modular architecture, user interaction, and realistic implementation are its strongest points. It does not, however, have quantifiable measurements, experimental evaluation, or deeper comparative insights with more recent approaches like GANs. Professional editing is also necessary for the writing style and grammar. Although the work has potential for use in digital art and education, it requires significant improvement to satisfy IEEE publication requirements.

10. Recommendation: Your overall rating

Accept with Major Revision
